

Tayo Olofintuyi

tolofi152@gmail.com | Cell: (336) 365-1621 | github.com/TayoOlofintuyi | linkedin.com/in/tayo-olofintuyi/ | Charlotte, NC 28278

SUMMARY

I am a solutions-oriented researcher, developer, and recent Computer Science graduate from Appalachian State University. I am looking for opportunities to build on the skills I have learned and practiced while participating in various research projects and outside professional endeavors.

SKILLS

- **Programming Languages:** Python, Java, C, C++, C#, SQL, HTML, JavaScript, Typescript, CSS, Arduino C, R, Assembly (LC3, Y86, X86), Haskell, Kotlin
- **Languages:** English (native), Spanish (intermediate, 1600+ Day Duolingo Streak), Yoruba (beginner)
- **Soft Skills:** Leadership, Mentorship, Collaboration, Communication, Problem-Solving, Conflict Resolution
- **Related Skills:** PC Based Hardware Basics, Database Management, QA & Testing, Troubleshooting, Game Design and Development (Unity, GameMaker, Godot, MIT App Inventor), Arduino, Angular, GDB, Pair Programming

EDUCATION

Appalachian State University – Boone, NC (August 2021 – December 2024)

- Bachelor of Science, Computer Science | GPA: 3.38

Awards and Scholarships

- Computer Science Undergraduate Leadership Award
- Louise Stokes Alliance for Minority Participation (LSAMP)
- ECRS CS High Achievers Scholarship
- Computer Science Alumni Scholarship
- The Camelot Scholarship for Computer Science
- FND-Goodman Math/Computer Science
- Chancellors and Deans List 3/8 semesters

EXPERIENCE

Ferrous Design Inc., Programmer and Contractor, (June 2024 – April 2025)

- Developed and managed security notification systems for client companies, enhancing real-time monitoring and response capabilities.
- Implemented automation using Slack API, Python (JSON package), and cron jobs to track and report entrance activity, improving efficiency and reliability.

Appalachian State Multipurpose Apiary Informatic Systems (AppMAIS), Research Assistant, (Oct 2022 – Dec 2024)

- Utilized machine learning and bioinformatics tools to investigate the health, development, and genomic diversity of honey beehives in North Carolina to understand and optimally prevent honey bee colony collapse disorder (CCD).
- Modified the AppMAIS codebase for efficient interaction with data using Python, MongoDB, and DataGrip.
- **Related Projects:**
 - **YOLOv8 for Determining Honey Beehive Health Based on the Cast Ratio** (May 2023 – December 2024)
 - Developed a machine learning model using PyTorch, Ultralytics YOLOv8, NumPy, and Matplotlib to analyze honey beehive health by tracking male-to-female ratios.
 - Co-authored a research paper (in progress for publication) based on project findings.
 - Presented results at the **2023 State of North Carolina Undergraduate Research and Creativity Symposium (SNCURCS)** at Wingate University.
 - **Stripe Segmentation** (Aug 2023 – Dec 2024)
 - Applied PyTorch, Ultralytics YOLOv8, NumPy, and Matplotlib to develop pipelines for data preprocessing, training, clustering, and visualization.
 - Utilized clustering algorithms like DBSCAN, Agglomerative Clustering, and Affinity Propagation to analyze honeybee stripe patterns as indicators of biodiversity.
 - Monitored model performance using WandB and TensorBoard, ensuring effective training and evaluation.
 - Reviewed academic literature to enhance understanding of clustering techniques and evaluation metrics.

University Housing, Resident Assistant, (August 2022 – May 2023)

- Implemented and enforced all college and campus living policies, promoting academic excellence, addressing student needs, and encouraging involvement.
- Created, marketed, and led student-centered hall programs to promote social, educational, diversity, health, emotional understanding, and growth.

CAMPUS LEADERSHIP AND INVOLVEMENT

STEM Seminar, *Scholar and Mentor* (January 2023 – December 2024)

- Participated in seminars and study halls focused on mentorship and research within minority STEM communities.
- Collaborated on projects with fellow scholars under the guidance of STEM faculty mentors.
- Provided peer mentorship in Python programming and software development.
- Presented research at the Mountain2Sea LSAMP Conference alongside scholars and faculty from other universities.
- **Related Projects:**
 - **The Journey Website** (Aug 2023 – Dec 2024)
 - Created a mental health support website with features for journaling, event tracking, emotion logging, and local resource access using HTML, CSS, JavaScript, Node.js, and MongoDB.
 - Focused on user-friendly design to improve accessibility and provide tailored resources for the Boone community.
 - **Course Relief** (Jan 2024 – May 2024)
 - Developed a website to streamline academic advising for computer science students, simplifying schedule building with HTML, CSS, JavaScript, and JSON.
 - Designed a data structure to assess course difficulty and workload, enhancing decision-making for students and advisors.

Women In Computer Science Club (WICS), *Publicist* (April 2022 – January 2024)

- Organized events to encourage bonding with and uplifting women and LGBTQ+ people in computer science.
- Maintained the club's social media pages, took pictures at events, and advertised for the club.

AppHack, *Director and Participant* (April 2022 – April 2023)

- Organized and supervised a hackathon where students could enhance their software development by creating projects.
- Collaborated with faculty and companies to partner with the director team in funding and presenting.

Artificial Intelligence/Machine Learning Club, *Participant* (May 2023 – September 2024)

- Audited Coursera courses from the Deep Learning Specialization taught by Andrew Ng and used knowledge to contribute to multiple research projects.

Henderson Springs LGBTQ Center, *Volunteer* (August 2021 – August 2023)

- Operated the front desk as the first face of introduction to the center.
- Organized events such as National Coming Out Day and Queer Prom on campus with fellow volunteers.

Justified Referral in AI Glaucoma Screening Challenge (JustRAIGS), *Participant* (January 2024 – April 2024)

- Analyzed a dataset of 110,000+ eye images to classify cases as referable and non-referable for glaucoma and present symptoms using AI models.
- Performed binary and multi-label classification to support the development of algorithms for early glaucoma detection.
- Gained experience with medical imaging datasets, machine learning workflows, and feature annotation techniques.

North Carolina Educational Datathon 2024, *Participant* (December 2023 – February 2024)

- Engaged with large amounts of data through data cleaning and analysis to discover trends and gaps between Career and Technical Education (CTE) and North Carolina workforce needs.
- Developed solutions for optimizing North Carolina K–12 students for the workforce.

OTHER PROJECTS AND EXPERIENCE

Capstone: Conundrum, RogueLike Game (August 2024 – December 2024)

- Developed Conundrum, a Roguelike game using GameMaker Studio and GML, incorporating combat, movement, and level progression mechanics.
- Designed a projectile system enabling intuitive aiming and shooting based on player movement and mouse input.
- Created custom pixel art assets using Piskart and Piskel, enhancing the game's visual style.
- Streamlined game flow and map transitions with a centralized controller object.
- Conducted iterative playtesting with external feedback, refining difficulty levels and optimizing the player experience.

SinWaves (January 2024 – May 2024)

- Developed a Haskell library for creating music and manipulating sound properties such as pitch, octave, and speed, usable in GHCI sessions.
- Gained foundational knowledge of functional programming through project implementation in Haskell.

Minesweeper Documentation (August 2023 – December 2023)

- Developed user-friendly documentation for the popular computer game Minesweeper.
- Used languages such as HTML, JavaScript, and CSS.